

ANALYSTLAB AFRICA ARTIFICIAL INTELLIGENCE INTERNSHIP PROGRAM

Batch B (June 1 – August 1, 2026)

Week 1 Assignment

AI FUNDAMENTALS REPORT

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1. Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)

Artificial Intelligence (AI) is the broad field of computer science dedicated to creating intelligent systems that can simulate human cognitive abilities such as reasoning, learning, decision-making, problem-solving, perception, and language processing. These systems may function through rule-based programming or by learning from data, enabling them to adapt and improve their performance in carrying out complex tasks.

Machine Learning (ML) is a subset of Artificial Intelligence (AI) that focuses on developing algorithms and statistical models that enable computer systems to learn patterns and relationships from data without being explicitly programmed for every situation. By learning from experience, ML systems can make predictions, support decision-making, solve problems, and continuously improve their performance, allowing them to adapt to new information and generalize their knowledge to previously unseen data.

Deep Learning (DL) is a specialized subset of Machine Learning that uses multi-layered artificial neural networks to automatically learn complex patterns and features from large datasets. By processing data through interconnected layers, deep learning models can analyze unstructured data such as images, audio, and text, making them highly effective for tasks such as image recognition, speech processing, and natural language understanding.

2. Differences between AI, ML and DL

AI	ML	DL
Makes machines intelligent	Makes machines learn from data	Makes machines learn using deep neural networks
Can work with rules or learning	Learns patterns from data	Learns complex patterns automatically
Broadest field	Subset of AI	Subset of ML

AI	ML	DL
Requires more human guidance	Requires some human guidance	Requires the least human guidance
Works with various approaches	Uses statistical learning algorithms	Uses deep neural networks

3. Real-world applications:

Applications of Artificial Intelligence (AI)

Artificial Intelligence (AI) is applied across various sectors to automate tasks, support decision-making, and simulate human intelligence. Common applications include virtual assistants such as Siri, Alexa, and Google Assistant, GPS navigation systems, automated scheduling platforms, and game-playing systems such as chess engines that can make strategic decisions in complex environments.

Applications of Machine Learning (ML)

Machine Learning (ML) is widely used in data-driven applications where systems learn from historical data to make predictions and improve performance. Examples include email spam detection systems that classify unwanted messages, recommendation engines used by Netflix, YouTube, and Amazon to personalize user experiences, and fraud detection systems that identify suspicious financial transactions.

Applications of Deep Learning (DL)

Deep Learning (DL) is particularly effective for analyzing complex and unstructured data such as images, audio, and text. It is widely used in facial recognition, image classification, object detection, speech recognition, autonomous driving, medical image analysis, language translation, and generative AI systems capable of producing text, images, audio, and video content.

References

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- Mitchell, T. M. (1997). Machine Learning. McGraw-Hill Education.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning. MIT Press.